

Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Wednesday 14 May 2025

Morning (Time: 1 hour 30 minutes)	Paper reference	8PH0/01
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Physics

Advanced Subsidiary

PAPER 1: Core Physics I

You must have: Scientific calculator and ruler Data, Formulae and Relationships Booklet (enclosed)	Total Marks
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Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions in Sections A and B.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The total mark for this paper is 80.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*
- In the question marked with an **asterisk** (*), marks will be awarded for your ability to structure your answer logically, showing how the points that you make are related or follow on from each other where appropriate.

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- You are advised to show your working in calculations including units where appropriate.

Turn over ►

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SECTION A

Answer ALL questions.

All multiple choice questions must be answered with a cross in the box ☐ for the correct answer from A to D. If you change your mind about an answer, put a line through the box ☐ and then mark your new answer with a cross ☐.

- 1 A ball bearing falls from rest through oil.

Which of the following statements is correct during the initial motion of the ball bearing?

- ☐ A The resultant force on the ball bearing decreases.
- ☐ B The resultant force on the ball bearing increases.
- ☐ C The resultant force on the ball bearing decreases then increases.
- ☐ D The resultant force on the ball bearing stays constant.

(Total for Question 1 = 1 mark)

- 2 As the temperature of a negative temperature coefficient thermistor increases, the resistance of the thermistor changes.

Which of the following statements is correct?

- ☐ A Resistance decreases as lattice ions vibrate with decreased amplitude.
- ☐ B Resistance increases as lattice ions vibrate with increased amplitude.
- ☐ C Resistance decreases as the number of conduction electrons increases.
- ☐ D Resistance increases as the number of conduction electrons decreases.

(Total for Question 2 = 1 mark)

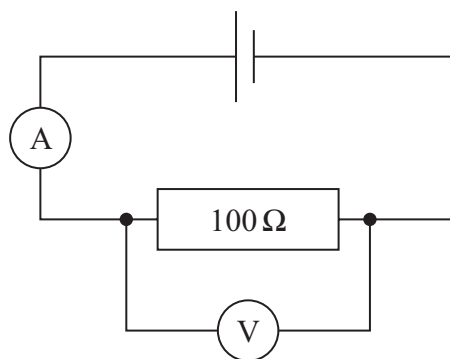
- 3 Which of the following units is equivalent to the newton?

- ☐ A kg m s^{-1}
- ☐ B kg m s^{-2}
- ☐ C $\text{kg m}^2 \text{s}^{-2}$
- ☐ D $\text{kg m}^2 \text{s}^{-1}$

(Total for Question 3 = 1 mark)



- 4 A circuit consists of a cell of e.m.f. 1.50 V connected to a $100\ \Omega$ resistor. A voltmeter is connected across the $100\ \Omega$ resistor, as shown. The voltmeter reading is 1.45 V .

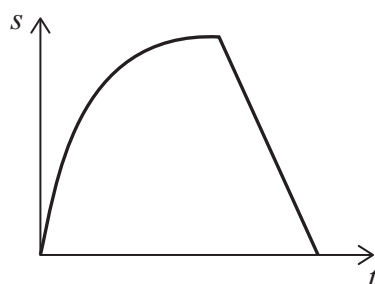


Which of the following gives the internal resistance, in Ω , of the cell?

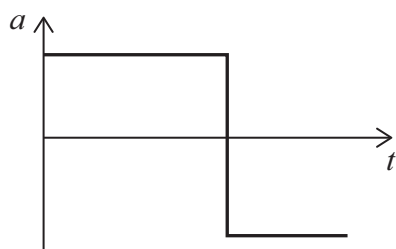
- ☐ A $\frac{(1.50 - 1.45) \times 100}{1.45}$
- ☐ B $\frac{1.50 \times 100}{1.45}$
- ☐ C $\frac{(1.50 - 1.45) \times 100}{1.50}$
- ☐ D $\frac{1.45 \times 100}{1.50}$

(Total for Question 4 = 1 mark)

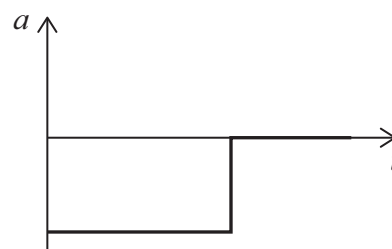
5 The graph shows how the displacement s varies with time t for an object.



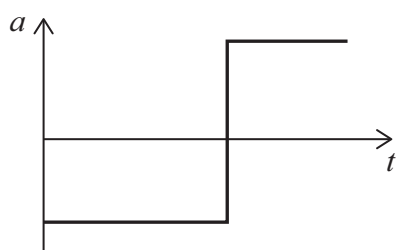
Which of the following graphs shows how acceleration a varies with time t for the object?



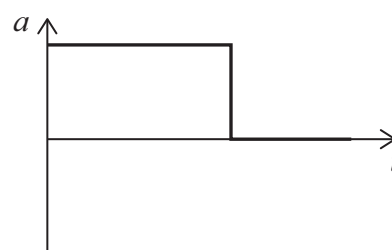
A



B



C



D

- ☐ A
- ☐ B
- ☐ C
- ☐ D

(Total for Question 5 = 1 mark)

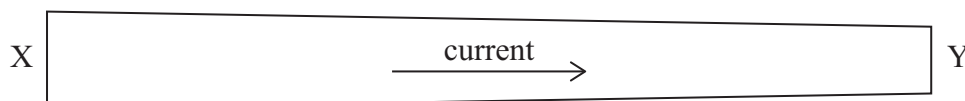
- 6 A stone is dropped from rest down a well. The sound as the stone hits the water is heard 3.5 s after the stone is dropped.

Which of the following gives the distance, in m, travelled by the stone before it hits the water?

- ☐ A $\frac{9.81 \times 3.5^2}{2}$
- ☐ B 9.81×3.5
- ☐ C $2 \times 9.81 \times 3.5^2$
- ☐ D $\frac{3.5^2 - 0}{2 \times 9.81}$

(Total for Question 6 = 1 mark)

- 7 A wire of decreasing diameter is connected in a circuit. There is a current from X to Y in the wire, as shown.



Which row of the table gives the current and the drift velocity of charge carriers at Y compared to X?

	Current	Drift velocity
☐ A	lower	higher
☐ B	lower	lower
☐ C	the same	higher
☐ D	the same	lower

(Total for Question 7 = 1 mark)

- 8 A shopping trolley moves at velocity v . The trolley collides and combines with a stack of 5 identical stationary trolleys.

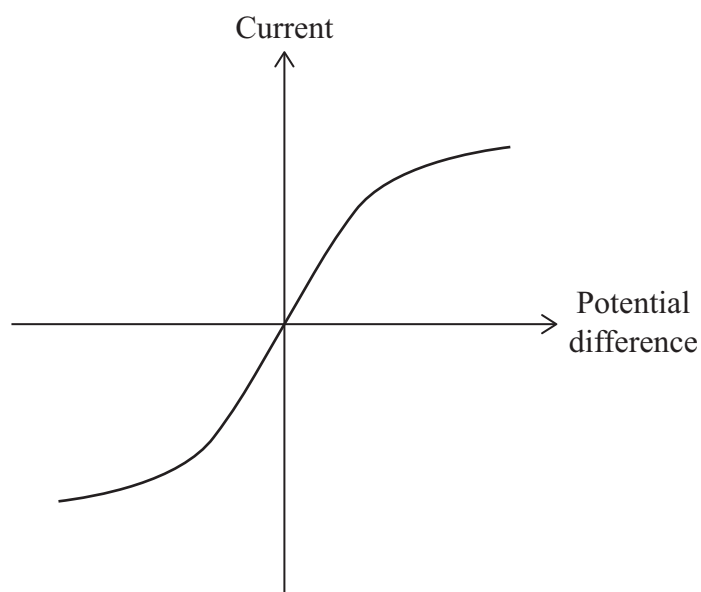
Which of the following expressions gives the velocity of the combined stack of trolleys?

- ☐ A $6v$
- ☐ B $5v$
- ☐ C $\frac{v}{5}$
- ☐ D $\frac{v}{6}$

(Total for Question 8 = 1 mark)



- 9 The graph shows how current varies with potential difference for a component in a circuit.



Explain how the graph shows what the component is.

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(Total for Question 9 = 3 marks)

10 A person pushes a baby in a pram.

- (a) The person pushes the pram with a constant force. The force is applied at an angle of 50° to the horizontal, as shown.



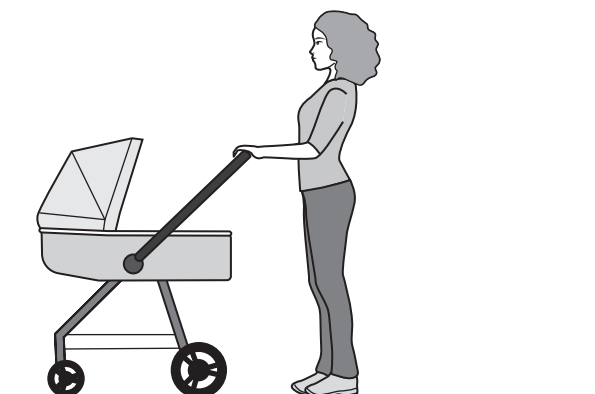
When a force of 35 N is applied, the pram moves at constant speed.

Calculate the work done to push the pram at constant speed for 500 m .

(3)

Work done =

- (b) The pram is then moved along the same surface, at the same constant speed, by a different person. This person pushes the pram by applying a force horizontally, as shown.

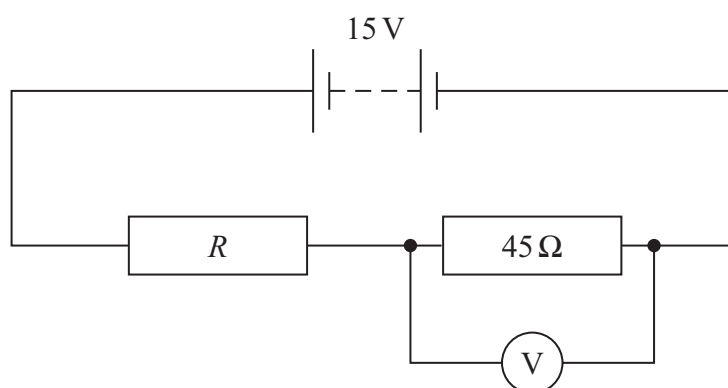


Explain how applying a force horizontally changes the magnitude of the force needed to push the pram at the same constant speed.

(2)

(Total for Question 10 = 5 marks)

- 11 Two resistors are connected in series in a circuit, as shown. The power supply has negligible internal resistance.



- (a) Calculate the resistance R of the resistor when the voltmeter reads 5.5 V.

(3)

$R = \dots\dots\dots$

- (b) A student uses the circuit to power an electric motor. He connects the motor in parallel with the $45\ \Omega$ resistor.

The electric motor has a resistance of $45\ \Omega$ and a normal working potential difference of 5.5 V.

Deduce how the motor performs when connected to the circuit. You do not need to do a further calculation.

(3)

(Total for Question 11 = 6 marks)



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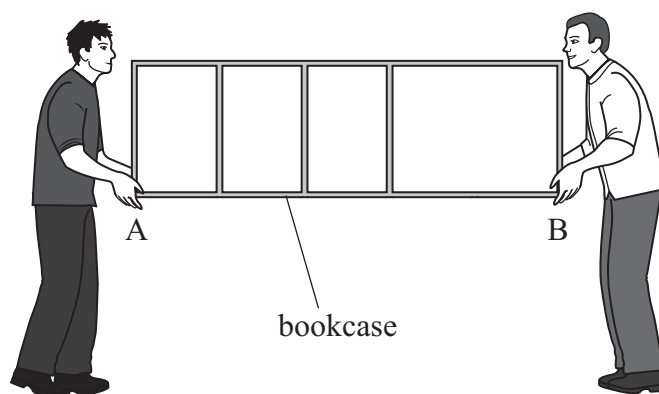
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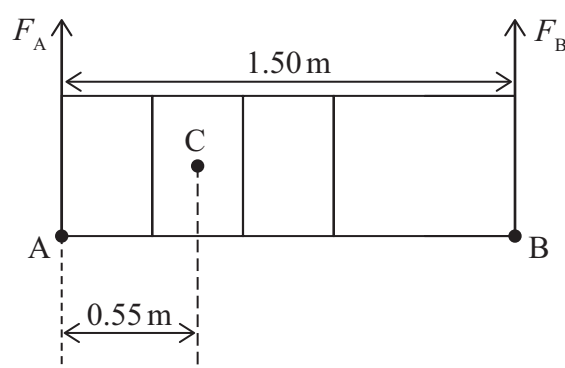
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- 12 Two people are carrying a bookcase. They are holding the bookcase at points A and B, as shown.



- (a) The length of the bookcase is 1.50 m. The centre of gravity C of the bookcase is 0.55 m from one end. The people exert forces F_A and F_B on the bookcase at points A and B, as shown below.



Calculate F_A and F_B .

mass of bookcase = 25 kg

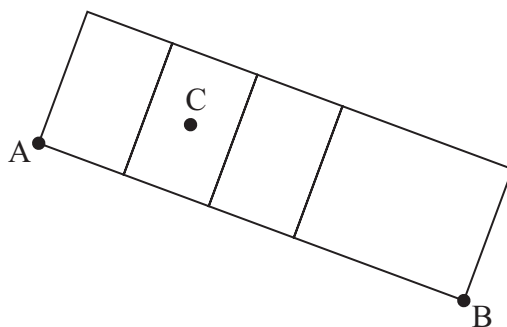
(5)

$$F_A = \dots\dots\dots$$

$$F_B = \dots\dots\dots$$



- (b) The two people now support the bookcase so that it is tilted with point A higher than point B, as shown. The bookcase is stationary and the forces exerted at points A and B are still vertical.



Explain how the magnitude of the forces exerted at points A and B change. You should consider moments about point A.

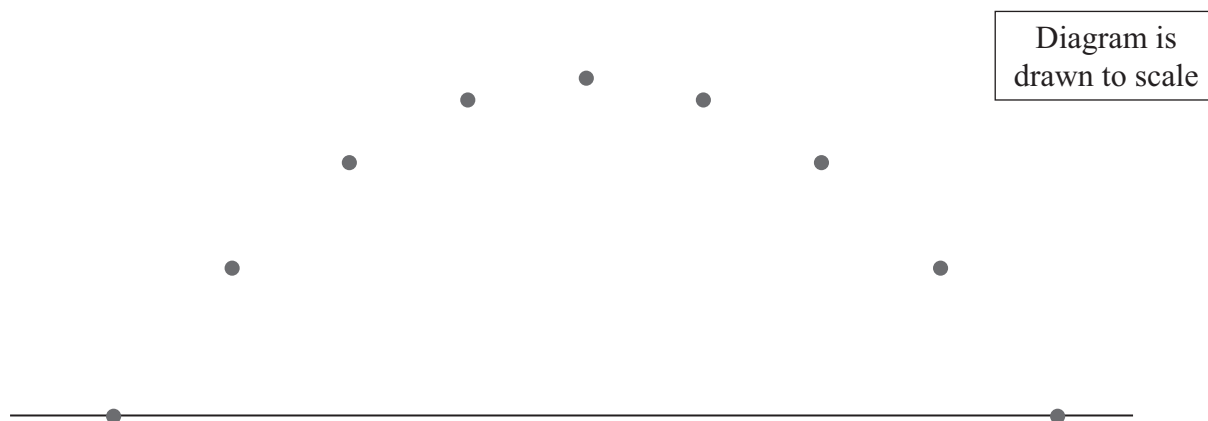
(5)

(Total for Question 12 = 10 marks)

- 13 A textbook has the following statement about the motion of a projectile with negligible air resistance acting on it.

‘The only force acting on a projectile is the weight of the projectile. This affects the vertical motion of the projectile but not the horizontal motion.’

A ball was projected up from the ground at an angle. The position of the ball was plotted at equal time intervals, as shown.



- *(a) Explain how the diagram gives evidence to support the statement in the textbook.

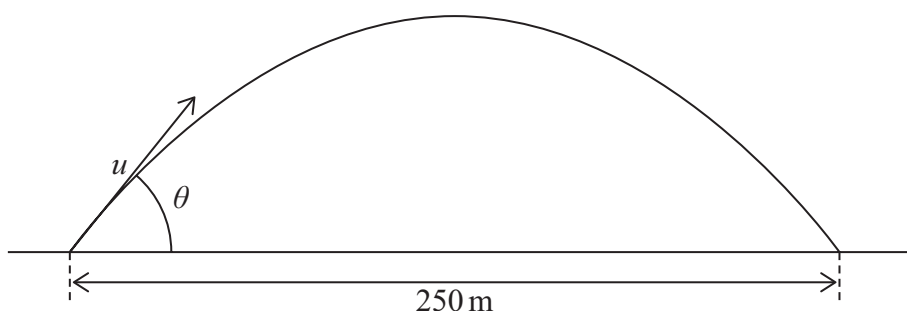
(6)

- (b) The photograph shows a cannon which was used to fire cannonballs over long distances in the 15th century.



(Source: © Ton Bangkeaw/Shutterstock)

A cannonball was projected over a horizontal distance of 250 m, as shown below. The time of flight of the cannonball was 7.5 s.



Calculate the angle θ at which the cannonball was fired and the initial speed u of the cannonball.

(5)

$$\theta = \dots\dots\dots$$

$$u = \dots\dots\dots$$

(Total for Question 13 = 11 marks)

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14 Since 2008, the Scottish island of Eigg has generated almost all its electrical power from renewable energy sources. The main source of power for the island is produced by hydroelectric generators.

(a) Water falls through a height of 124 m as it flows through a pipe to a hydroelectric generator.

(i) Calculate the maximum power input of the generator.

mass flow rate of the water = 130 kg s^{-1}

(2)

Maximum power input =

(ii) Explain why the power input of the generator is always less than this value.

(2)



- (b) When demand for power is low on Eigg, excess energy is stored in a number of batteries. The batteries have a total capacity of 4400 Ah when fully charged.

When demand for power is greater than the output from the renewable sources, energy is supplied by the batteries. The batteries supply an average power of 20 kW.

Calculate the total time, in hours, for which the batteries can supply energy before they need recharging.

total output potential difference of batteries = 48 V

(3)

Total time = hours

- (c) Each house on Eigg is allowed a maximum power of 5 kW.

- (i) Show that the maximum current in an electrical cable to a house is about 22 A.

mains potential difference = 230 V

(2)



(ii) On Eigg the cables to the houses are made of aluminium instead of copper.

The rate of energy transfer to the environment is limited to 0.05 W per metre of cable.

The engineers claimed that using aluminium instead of copper reduced the cost of the electrical cables by 90%.

Evaluate this claim by comparing a 1 m length of cable made from each material.

Assume the current in each cable is 22 A.

Material	Resistivity / $\Omega \text{ m}$	Cost in £ per m^3
Copper	1.68×10^{-8}	77 090
Aluminium	2.65×10^{-8}	6070

(5)

(Total for Question 14 = 14 marks)

TOTAL FOR SECTION A = 57 MARKS



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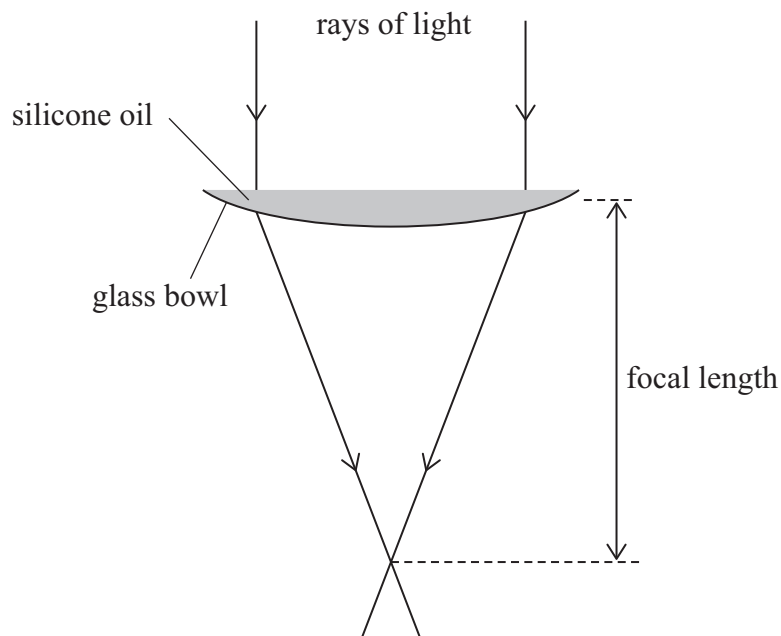


SECTION B

Answer ALL questions in the spaces provided.

- 15 A student makes a lens by filling a glass bowl with silicone oil. She investigated the refraction of light through this lens.

The student shone light vertically downwards through the lens, as shown.



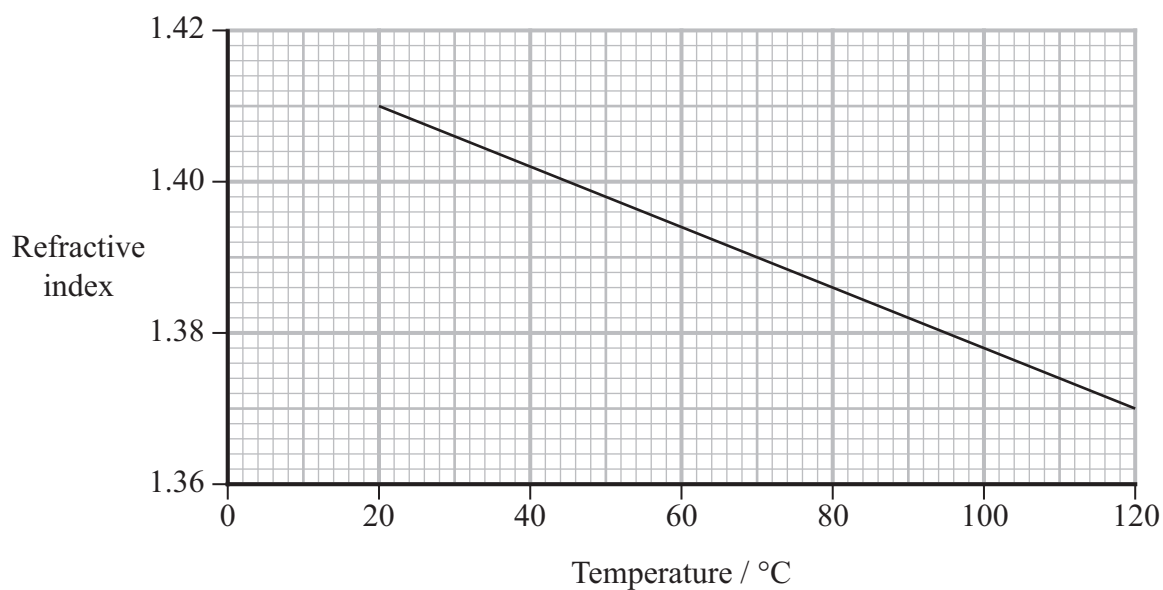
- (a) The student measured the focal length of the lens to be 16.7 cm.

Calculate the power of the lens.

(2)

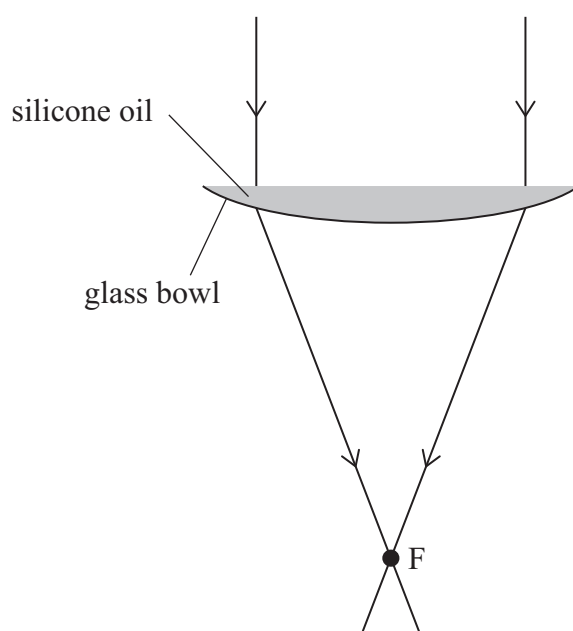
Power =

(b) The refractive index of the silicone oil changes with temperature, as shown.



The student connected a light dependent resistor (LDR) to an ohmmeter.

The light was brought to a focus at the focal point F, as shown. The LDR was placed at F.



The student recorded the resistance of the LDR. She then filled the glass bowl with silicone oil at a higher temperature and recorded the resistance of the LDR again.

Deduce the effect on the resistance of the LDR of increasing the temperature of the oil.

(5)

(Total for Question 15 = 7 marks)

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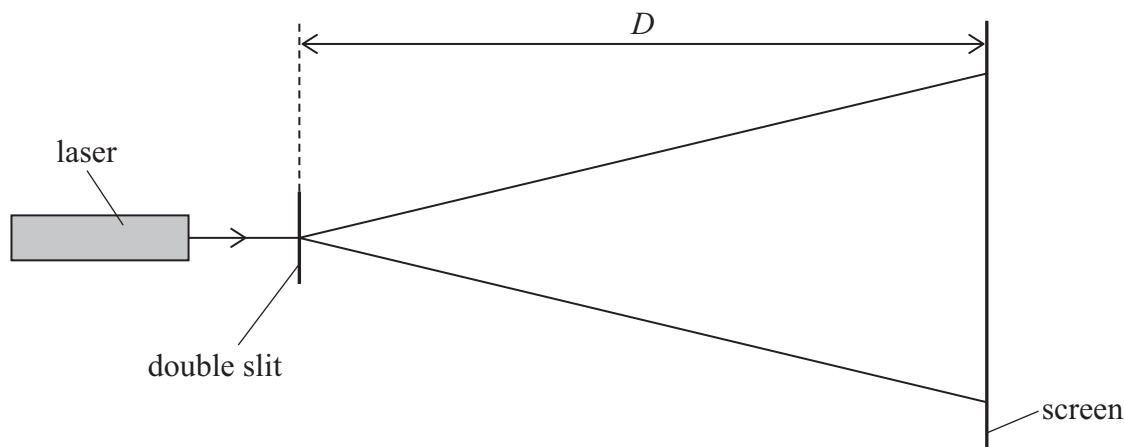
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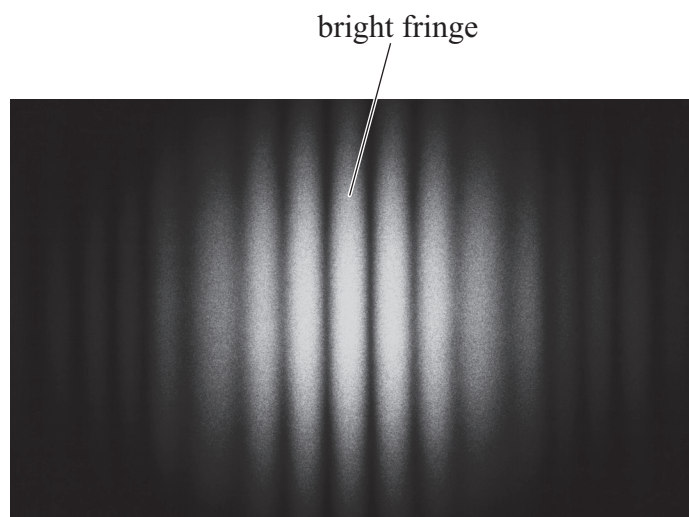
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- 16 A student directed light from a laser onto a double slit. The double slit consisted of two parallel narrow slits a distance s apart. The light spread out as it passed through the double slit and hit a screen a distance D away from the double slit, as shown.



The light produced a pattern of bright and dark fringes on the screen, as shown below.



(Source: © GIPhotoStock/SCIENCE PHOTO LIBRARY)

- (a) The student used digital calipers to measure the separation of the fringes w by measuring between the centres of adjacent fringes.

Explain how the student could reduce the percentage uncertainty in his measurement of w .

(2)

.....

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- (b) The student varied s for the double slit and recorded corresponding values of w . The results are recorded below.

s / mm	w / m	
0.10	0.0643	
0.20	0.0301	
0.30	0.0204	
0.40	0.0159	
0.50	0.0131	

- (i) Plot a graph of s on the y -axis and $1/w$ on the x -axis using the grid opposite.

Use the additional column in the table for your processed data.

(5)

- (ii) The following equation can be used to determine the wavelength λ of laser light for the experiment.

$$s = \frac{\lambda D}{w}$$

Calculate λ for the light used in this experiment.

$$D = 9.00 \text{ m}$$

(3)

$$\lambda = \dots\dots\dots$$



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(c) Another student carried out the same experiment using a different laser labelled with a wavelength of 520 nm.

- (i) She measured w several times for the same value of s . The results are recorded below.

w / mm				$w_{\text{mean}} / \text{mm}$
21	20	23	24	22

Calculate the percentage uncertainty in the value of w .

(2)

Percentage uncertainty in $w = \dots\dots\dots$

- (ii) Assess whether the student's results support the value of the wavelength of light given on the label.

$$s = \frac{\lambda D}{w}$$

$$D = 9.00 \text{ m} \pm 0.01 \text{ m}$$

$$s = 0.20 \text{ mm} \pm 4\%$$

(4)

(Total for Question 16 = 16 marks)

TOTAL FOR SECTION B = 23 MARKS
TOTAL FOR PAPER = 80 MARKS



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reference**

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Physics

Advanced Subsidiary

PAPER 1: Core Physics I

Data, Formulae and Relationships Booklet

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List of data, formulae and relationships

Acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$	(close to Earth's surface)
Electron charge	$e = -1.60 \times 10^{-19} \text{ C}$	
Electron mass	$m_e = 9.11 \times 10^{-31} \text{ kg}$	
Electronvolt	$1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$	
Gravitational field strength	$g = 9.81 \text{ N kg}^{-1}$	(close to Earth's surface)
Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$	
Speed of light in a vacuum	$c = 3.00 \times 10^8 \text{ m s}^{-1}$	

Mechanics

Kinematic equations of motion

$$s = \frac{(u + v)t}{2}$$

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

Forces

$$\Sigma F = ma$$

$$g = \frac{F}{m}$$

$$W = mg$$

$$\text{moment of force} = Fx$$

Momentum

$$p = mv$$

Work, energy and power

$$\Delta W = F\Delta s$$

$$E_k = \frac{1}{2}mv^2$$

$$\Delta E_{\text{grav}} = mg\Delta h$$

$$P = \frac{E}{t}$$

$$P = \frac{W}{t}$$

$$\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy input}}$$

$$\text{efficiency} = \frac{\text{useful power output}}{\text{total power input}}$$

Electricity

Potential difference

$$V = \frac{W}{Q}$$

Resistance

$$R = \frac{V}{I}$$

Electrical power and energy

$$P = VI$$

$$P = I^2R$$

$$P = \frac{V^2}{R}$$

$$W = VIt$$

Resistivity

$$R = \frac{\rho l}{A}$$

Current

$$I = \frac{\Delta Q}{\Delta t}$$

$$I = nqvA$$



Materials

Density

$$\rho = \frac{m}{V}$$

Stokes' law

$$F = 6\pi\eta r v$$

Hooke's law

$$\Delta F = k\Delta x$$

Young modulus

$$\text{Stress } \sigma = \frac{F}{A}$$

$$\text{Strain } \varepsilon = \frac{\Delta x}{x}$$

$$E = \frac{\sigma}{\varepsilon}$$

Elastic strain energy

$$\Delta E_{\text{el}} = \frac{1}{2} F \Delta x$$

Waves and particle nature of light

Wave speed

$$v = f\lambda$$

Speed of a transverse wave on a string

$$v = \sqrt{\frac{T}{\mu}}$$

Intensity of radiation

$$I = \frac{P}{A}$$

Power of a lens

$$P = \frac{1}{f}$$

$$P = P_1 + P_2 + P_3 + \dots$$

Thin lens equation

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

Magnification for a lens

$$m = \frac{\text{image height}}{\text{object height}} = \frac{v}{u}$$

Diffraction grating

$$n\lambda = d \sin \theta$$

Refractive index

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$n = \frac{c}{v}$$

Critical angle

$$\sin C = \frac{1}{n}$$

Photon model

$$E = hf$$

Einstein's photoelectric equation

$$hf = \phi + \frac{1}{2}mv_{\text{max}}^2$$

de Broglie wavelength

$$\lambda = \frac{h}{p}$$

END OF DATA, FORMULAE AND RELATIONSHIPS LIST



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